

UNIT 3 – INDEX NUMBER

Introduction:

In business, managers and other decision makers may be concerned with the way in which

the values of variables change over time like prices paid for raw materials, numbers of employees and customers, annual income and profits, and so on. Index numbers are one way of describing such changes.

If we turn to any journal devoted to economic and financial matters, we are very likely to come across an index number of one or the other type. It may be an index number of share prices or a wholesale price index or a consumer price index or an index of industrial production. The objective of these index numbers is to measure the changes that have occurred in prices, cost of living, production, and so forth. *For example*, if a wholesale price index number for the year 2000 with base year 1990 was 170; it shows that wholesale prices, in general, increased by 70 percent in 2000 as compared to those in 1990. Now, if the same index number moves to 180 in 2001, it shows that there has been 80 percent increase in wholesale prices in 2001 as compared to those in 1990.

With the help of various index numbers, economists and businessmen are able to describe and appreciate business and economic situations quantitatively. Index numbers were originally developed by economists for monitoring and comparing different groups of goods. It is necessary in business to understand and manipulate the different published index serieses, and to construct index series of your own. Having constructed your own index, it can then be compared to a national one such as the RPI, a similar index for your industry as a whole and also to indexes for your competitors. These comparisons are a very useful tool for decision making in business.

WHAT ARE INDEX NUMBERS?

“Index numbers are statistical devices designed to measure the relative changes in the level of a certain phenomenon in two or more situations”. The phenomenon under consideration may be any field of quantitative measurements. It may refer to a single variable or a group of distinct but related variables. In Business and Economics, the phenomenon under consideration may be:

the prices of a particular commodity like steel, gold, leather, *etc.* or a group of commodities like consumer goods, cereals, milk and milk products, cosmetics, *etc.*

volume of trade, factory production, industrial or agricultural production, imports or exports, stocks and shares, sales and profits of a business house and so on. the national income of a country, wage structure of workers in various sectors, bank deposits, foreign exchange reserves, cost of living of persons of a particular community, class or profession and so on.

The various situations requiring comparison may refer to either

the changes occurring over a time, or

the difference(s) between two or more places, or

the variations between similar categories of objects/subjects, such as persons, groups of persons, organisations etc. or other characteristics such as income, profession, etc.

The utility of index numbers in facilitating comparison may be seen when, *for example* we are interested in studying the general change in the price level of consumer goods, *i.e.* good or commodities consumed by the people belonging to a particular section of society, say, low income group or middle income group or labour class and so on. Obviously these changes are not directly measurable as the price quotations of the various commodities are available in different units, *e.g.*, cereals (wheat, rice, pulses, etc) are quoted in Rs per quintal or kg; water in Rs per gallon; milk, petrol, kerosene, etc. in Rs per liter; cloth in Rs per meter and so on.

Further, the prices of some of the commodities may be increasing while those of others may be decreasing during the two periods and the rates of increase or decrease may be different for different commodities. Index number is a statistical device, which enables us to arrive at a single representative figure that gives the general level of the price of the phenomenon (commodities) in an extensive group. According to Wheldon:

“Index number is a statistical device for indicating the relative movements of the data where measurement of actual movements is difficult or incapable of being made.”

Uses or Applications of Index Number:

The first index number was constructed by an Italian, Mr G R Carli, in 1764 to compare the changes in price for the year 1750 (current year) with the price level in 1500 (base year) in order to study the effect of discovery of America on the price level in Italy. Though originally designed to study the general level of prices or accordingly purchasing power of money, today index numbers are extensively used for a variety of purposes in economics, business, management, etc., and for quantitative data relating to production, consumption, profits, personnel and financial matters etc., for comparing changes in the level of phenomenon for two periods, places, etc. In fact there is hardly any field or quantitative measurements where index numbers are not constructed. They are used in almost all sciences – natural, social and physical. The main uses of index numbers can be summarized as follows:

1. Index Numbers as Economic Barometers

Index numbers are indispensable tools for the management personnel of any government organisation or individual business concern and in business planning and formulation of executive decisions. The indices of prices (wholesale & retail), output (volume of trade, import and export, industrial and agricultural production) and bank deposits, foreign exchange and reserves etc., throw light on the nature of, and variation in the general economic and business activity of the country. They are the indicators of business environment. A careful study of these indices gives

us a fairly good appraisal of the general trade, economic development and business activity of the country.

2. Index Numbers Help in Studying Trends and Tendencies

Since the index numbers study the relative change in the level of a phenomenon at different periods of time, they are especially useful for the study of the general trend for a group phenomenon in time series data. The indices of output (industrial and agricultural production), volume of trade, import and export, etc., are extremely useful for studying the changes in the level of phenomenon due to the various components of a time series, viz. secular trend, seasonal and cyclical variations and irregular components and reflect upon the general trend of production and business activity. As a measure of average change in extensive group, the index numbers can be used to forecast future events. For instance, if a businessman is interested in establishing a new undertaking, the study of the trend of changes in the prices, wages and incomes in different industries is extremely helpful to him to frame a general idea of the comparative courses, which the future holds for different undertakings.

3. Index Numbers Help in Formulating Decisions and Policies

Index numbers of the data relating to various business and economic variables serve an important guide to the formulation of appropriate policy. *For example*, the cost of living index numbers are used by the government and, the industrial and business concerns for the regulation of dearness allowance (D.A.) or grant of bonus to the workers so as to enable them to meet the increased cost of living from time to time. The excise duty on the production or sales of a commodity is regulated according to the index numbers of the consumption of the commodity from time to time. Similarly, the indices of consumption of various commodities help in the planning of their future production. Although index numbers are now widely used to study the general economic and business conditions of the society, they are also applied with advantage by sociologists (population indices), psychologists (IQs'), health and educational authorities etc., for formulating and revising their policies from time to time.

4. Price Indices Measure the Purchasing Power of Money

A traditional use of index numbers is in measuring the purchasing power of money. Since the changes in prices and purchasing power of money are inversely related, an increase in the general price index indicates that the purchasing power of money has gone down.

In general, the purchasing power of money may be computed as

$$\text{Purchasing Power} = \frac{1}{\text{General Price Index}} \times 100$$

Accordingly, if the consumer price index for a given year is 150, the purchasing power of a rupee is $(1/150) 100 = 0.67$. That is, the purchasing power of a rupee in the given year is 67 paise as compared to the base year.

With the increase in prices, the amount of goods and services which money wages can buy (or the real wages) goes on decreasing. Index numbers tell us the change in real wages, which are obtained as

$$\text{Real Wage} = \frac{\text{Money Wage}}{\text{Consumer Price Index}} \times 100$$

A real wage index equal to, say, 120 corresponding to money wage index of 160 will indicate an increase in real wages by only 20 per cent as against 60 per cent increase in money wages.

Index numbers also serve as the basis of determining the terms of exchange. The terms of exchange are the parity rate at which one set of commodities is exchanged for another set of commodities. It is determined by taking the ratio of the price index for the two groups of commodities and expressing it in percentage.

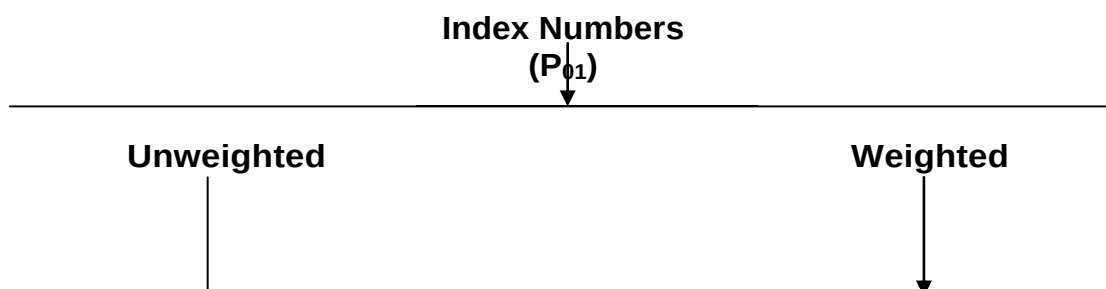
For example, if A and B are the two groups of commodities with 120 and 150 as their price index in a particular year, respectively, the ratio $120/150$ multiplied by 100 is 80 per cent. It means that prices of A group of commodities in terms of those in group B are lower by 20 per cent.

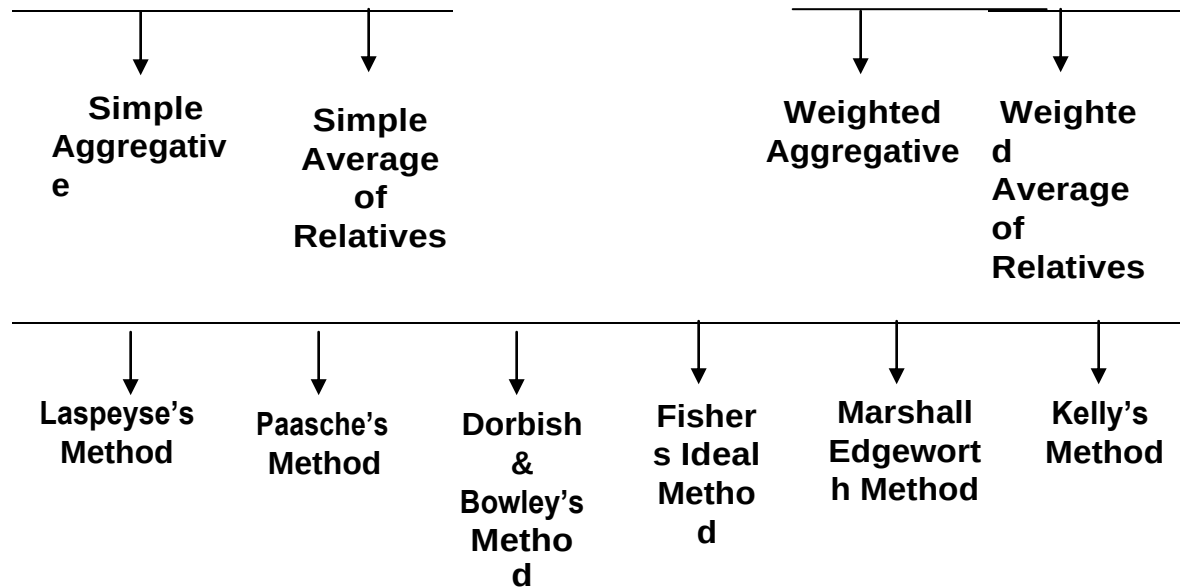
5. Index Numbers are Used for Deflation

Consumer price indices or cost of living index numbers are used for deflation of net national product, income value series in national accounts. The technique of obtaining real wages from the given nominal wages (as explained in use 4 above) can be used to find real income from inflated money income, real sales from nominal sales and so on by taking into account appropriate index numbers.

6. Universal Application : Index numbers are useful in economic, commercial, social and in every field such as agriculture etc.

Explain the various methods of constructing Index Numbers?





1. Unweighted Index Number:

(a) Simple Aggregative Method :

$$P_{01} = \frac{\sum P_1 \times 100}{\sum P_0}$$

Where –

$\sum P_1$ = Total of current year prices of various commodities.

$\sum P_0$ = Total of base year prices of various commodities.

(b) Simple Average of Relative Method :

$$P_{01} = \frac{\sum R}{N} \quad \text{and} \quad R = \frac{P_1 \times 100}{P_0}$$

Where –

P_1 = Prices of current year.

P_0 = Prices of base year.

Base year can be price of one year or multi-year or average price. Geometric mean

can also be used for averaging the price relatives.

2. Weighted Index Numbers:

(a) Weighted Aggregative Method:

(i) Lapser's Method

$$P_{01} = \frac{\sum P_1 q_0}{\sum P_0 q_0} \times 100$$

Where –

$P_1 \rightarrow$ Price of current year.
 $P_0 \rightarrow$ Price of base year.
 $q_0 \rightarrow$ Quantity of base year

(ii) Paasche's Method:

$$P_{01} = \frac{\sum P_1 q_1}{\sum P_0 q_1} \times 100$$

Where –

$q_1 \rightarrow$ Quantity of current year.

(iii) Dorbish & Bowley's Method:

$$P_{01} = \frac{\text{Laspeyre's Index} + \text{Paasche's Index}}{2}$$

OR

$$\left(\frac{\frac{\sum P_1 q_0 + \sum P_1 q_1}{\sum P_0 q_0 + \sum P_0 q_1}}{2} \right) \times 100$$

(IV) Marshall - Edge worth's Method:

$$P_{01} = \left(\frac{\frac{\sum P_1 q_0 + \sum P_1 q_1}{\sum P_0 q_0 + \sum P_0 q_1}}{2} \right) \times 100$$

(V) Fisher's Ideal Method:

$$P_{01} = \sqrt{\left(\frac{\sum P_1 q_0}{\sum P_0 q_0} \times \frac{\sum P_1 q_1}{\sum P_0 q_1} \right)} \times 100 \quad \text{or} \quad \sqrt{L \times P}$$

(VI) Kelly's Method:

$$\frac{\sum P_1 q}{\sum P_0 q} \times 100$$

Where:

$$q = q_0 + q_1$$

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Fisher's Formula is called the Ideal Formula. Why?

Ans.: Fisher's formula is called the ideal because of the following reasons:

- (i) It is based on geometric mean which is considered best for constructing index numbers.
- (ii) It fulfills both the time reversal and factor reversal tests.
- (iii) It takes into account both current year as well as base year prices and quantities.
- (iv) It is free from bias.

What are the methods of constructing Consumer Price Index or Cost of Living Index Numbers?

Consumer price index or cost of living index numbers are designed to measure the effect of changes in the price of group of commodities and services on the purchasing power of a particular class of society, during any given period with reference to some fixed base. Its object is to find out how much the consumers of a particular class have to pay more for a certain basket of goods and services in a given period as compared to the base period.

Following two methods are used to construct consumer price index numbers

- (v) Family Budget method or weighted average of relatives method.
- (vi) Aggregative Expenditure Method.

Family Budget Method : In this method, the price relatives for each commodity are obtained and these price relatives are multiplied by the value weights for each item and the product is divided by the total of weights.

$$\text{Consumer price index} = \frac{\sum RW}{\sum W}$$

$$\text{Where } R = \frac{P_1}{P_0} \times 100 \text{ and } W = P_0 q_0$$

This method is same as weighted average of price relatives method.

Aggregative Expenditure Method : This method is most popular method for constructing consumer price index numbers. The prices of commodities of current year as well as base year are multiplied by the quantities consumed in the base year. The aggregate expenditure of current year is divided by the aggregate expenditure of the base year and the quotient is multiplied by 100.

$$\text{Consumer price Index} = \frac{\sum P_1 q_0}{\sum P_0 q_0} \times 100$$

This method is same as Laspeyre's method.